

week-6(Pravalika-2520030402)

Question:

Create a client-server application using Named Pipes (FIFOs) and create a POSIX signal handling program for **SIGINT**, **SIGTERM**, and **SIGUSR1**.

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <string.h>
#include <signal.h>

#define FIFO "myfifo"

void signal_handler(int sig)
{
    if (sig == SIGINT)
        printf("\nSIGINT received\n");

    else if (sig == SIGTERM)
        printf("\nSIGTERM received\n");

    else if (sig == SIGUSR1)
        printf("\nSIGUSR1 received\n");
}

int main(int argc, char *argv[])
{
    char buffer[100];

    signal(SIGINT, signal_handler);
    signal(SIGTERM, signal_handler);
    signal(SIGUSR1, signal_handler);

    if (argc < 2)
    {
        printf("Use: %s server OR %s client\n", argv[0], argv[0]);
        return 1;
    }

    if (strcmp(argv[1], "server") == 0)
    {
        mkfifo(FIFO, 0666);
```

```

printf("Server started. PID = %d\n", getpid());

int fd = open(FIFO, O_RDONLY);

while (1)
{
    int n = read(fd, buffer, sizeof(buffer) - 1);

    if (n <= 0)
        break;

    buffer[n] = '\0';

    printf("Client: %s\n", buffer);

    if (strcmp(buffer, "exit") == 0)
        break;
}

close(fd);
unlink(FIFO);

printf("Server stopped.\n");
}

else if (strcmp(argv[1], "client") == 0)
{
    printf("Client started. PID = %d\n", getpid());

    int fd = open(FIFO, O_WRONLY);

    while (1)
    {
        printf("Enter message: ");

        fgets(buffer, sizeof(buffer), stdin);

        buffer[strcspn(buffer, "\n")] = '\0';

        write(fd, buffer, strlen(buffer) + 1);

        if (strcmp(buffer, "exit") == 0)
            break;
    }

    close(fd);
}

```

```
        printf("Client stopped.\n");
    }

    else
    {
        printf("Invalid option.\n");
        printf("Use: %s server OR %s client\n", argv[0], argv[0]);
    }

    return 0;
}
```

Output:

```
rikki@BOOK-GJU4MP9458:~$ ./a.out server
Server started. PID = 1028
```